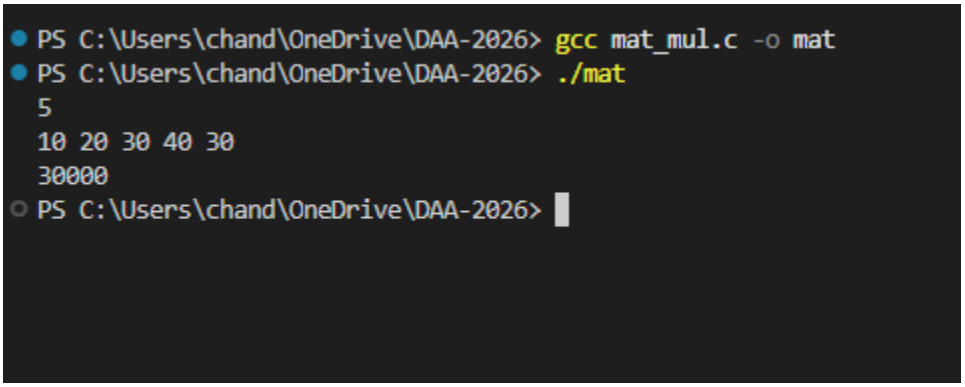
Mutchumari Chandrasekhar Reddy

AP25110010976

1.Implement Matrix Chain Multiplication using Dynamic Programming

Program:

```
#include<stdio.h>
 #include<limits.h>
int main(){
int n,i,j,k,l;
 scanf("%d",&n);
 int p[n];
 for(i=0;i<n;i++)
{
 scanf("%d",&p[i]);
 }
 int dp[n][n];
 for(i=1;i<n;i++)
{
dp[i][i]=0;
 }
 for(l=2;l<n;l++)
{
 for(i=1;i<n-l+1;i++)
{
j=i+l-1;
dp[i][j]=INT_MAX;
 for(k=i;k<j;k++){
int cost=dp[i][k]+dp[k+1][j]+p[i-1]*p[k]*p[j];
 if(cost<dp[i][j])
{
dp[i][j]=cost;
 }
 }
 }
```

```
}
printf("%d",dp[1][n-1]);
return 0;
}
```

Terminal Process: